

MODUL UNIVERSITY VIENNA

SEID 2364: Societal and Ethical Impact of Data Science

SESSION 2

Intelligence,
Consciousness,
Agency

David B. Smith



Opening Provocations

- Stream of consciousness.
- Capture your **immediate intuition**, not a polished or argued answer.
- Do not analyze or justify.
- Words or short phrases are enough.

(Write. Do not discuss yet.)



Opening
Provocations

What is “intelligence”?

(Write. Do not discuss yet.)

A graphic on the left side of the slide. It consists of a dark blue vertical rectangle. Overlaid on this rectangle is a lighter blue circle. The text "Opening Provocations" is written in white, sans-serif font, centered within the circle.

Opening
Provocations

How do you recognize
intelligence?

(Write. Do not discuss yet.)

Opening
Provocations

Is consciousness a prerequisite for
intelligence?

Why do you think so?

(Write. Do not discuss yet.)



Slide 6: Today's activities

- Provocations
- Administrative Onboarding
- Boundaries of Intelligence
- Systems & Decision
- Historical Theories of Intelligence
- Toward a Functional Definition
- Assignment

Slide 7: Administrative



GitHub

- Account created and login confirmed
- We will not build out repositories today



Zotero

- Account created
- Desktop app + browser connector installed
- Group library access confirmed



Ethical Journal

- Confirm you have created: “SEID 2364 – Ethical Journal”
- Cloud-based, editable, continuously updated
- Reliable access from your laptop
- Migration Migration to GitHub

Ethical Journal (required today) • Confirm you have created: “SEID 2364 – Ethical Journal” • Cloud-based, editable, continuously updated • Reliable access from your laptop • This is where you will record provocations and reflections •

Migration to GitHub will occur later — not today

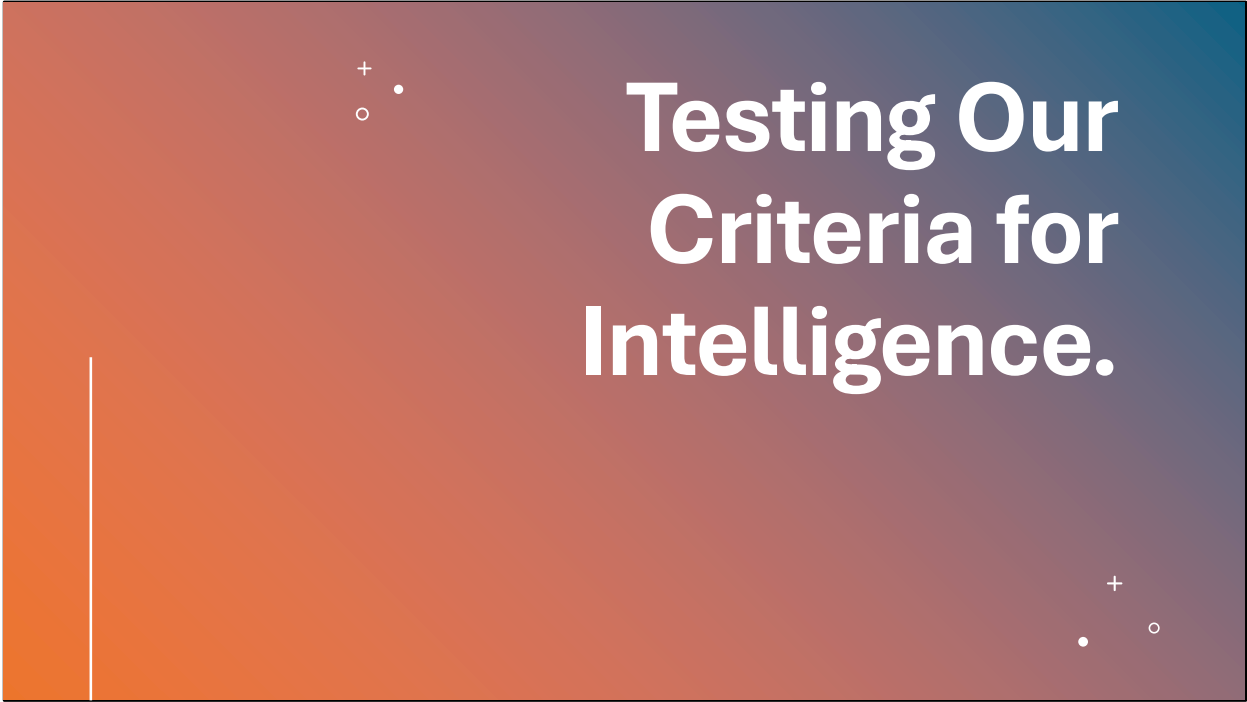
GitHub • Account created and login confirmed • We will not build out repositories today

Zotero • Account created • Desktop app + browser connector installed • Group library access confirmed



How do WE recognize intelligence?

- As a class, compile a shared list from our provocation.
- Capture all proposed conditions on the whiteboard.
- Discuss and refine as a class.
- This becomes our working test framework.
- Note that individuals may develop different opinions



Testing Our Criteria for Intelligence.

**Slide 9: Would
All Humans
Apply?**

**Testing Edge
Cases:**

- Infant
- Sleeping person
- Person in a coma

When did intelligence begin?





Lascaux Cave

Upper Paleolithic (~17,000 BCE)



Discovered September 10, 1940

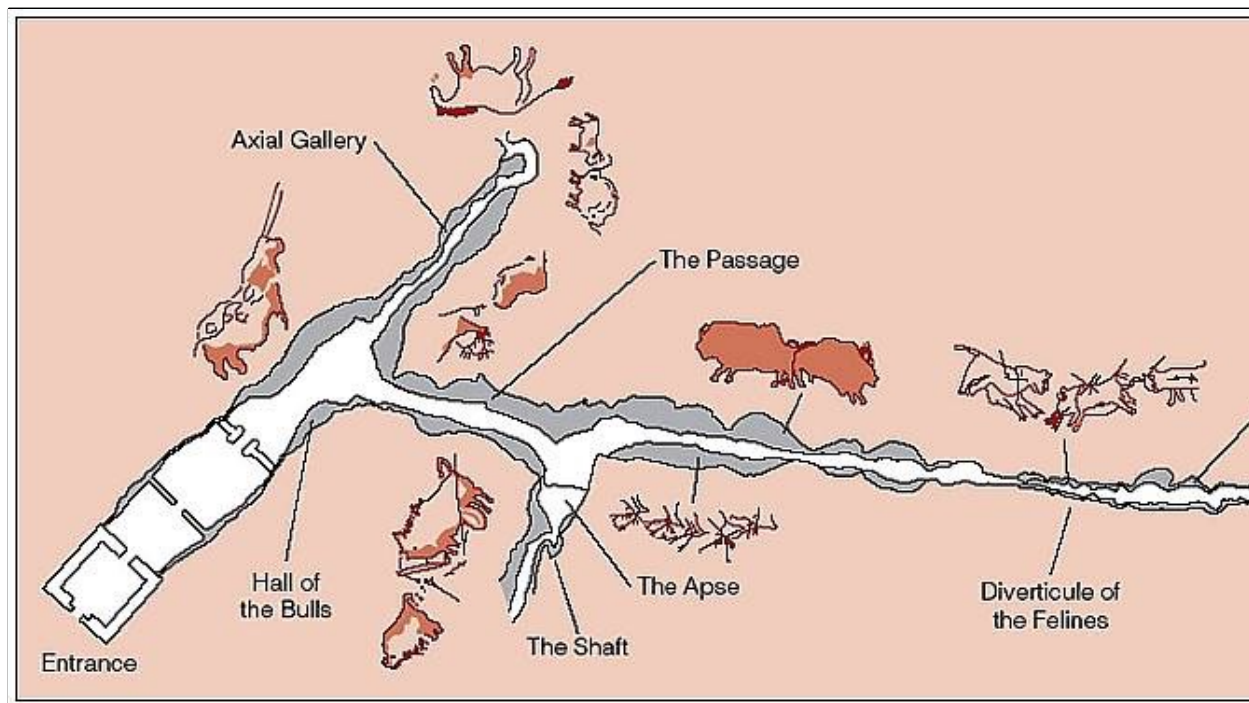


Four Boys looking for their dog.









To reach the Shaft of the Dead Man in Lascaux to see the only human figure (the bird-headed man), one must navigate to the deepest, most confined part of the cave complex. The shaft is located in the floor of the **Apse**, requiring a steep descent of approximately 17 to 19 meters below the main cave level, traditionally accessed via a ladder into a small, narrow fissure.

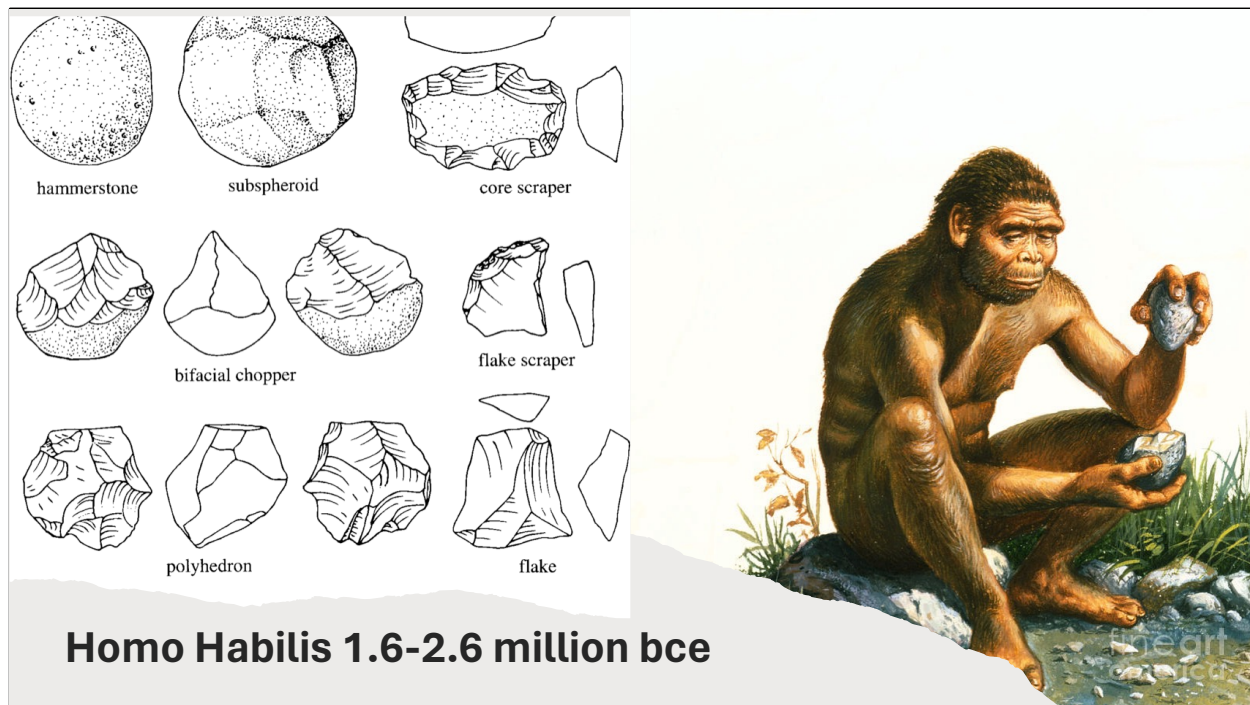


The Scene: The image, often called the "Bird Man" or "Bird-Headed Human Figure," depicts a bird-masked human with an ithyphallic condition (erect phallus) standing before a disemboweled, wounded bison.

The Location's Context: The Shaft is a deep, dangerous, and cramped natural fissure that leads to a lower level, situated approximately 6 meters below the main cave floor.

Other Imagery in the Shaft: Near the human figure is a bird-headed staff (or wand) and a rhinoceros that appears to be walking away.

Significance: Because it is in the deepest, most inaccessible part of the cave, some researchers believe this scene acts as a "sanctuary" or "holy of holies" relating to shamanic, magical, or storytelling purposes

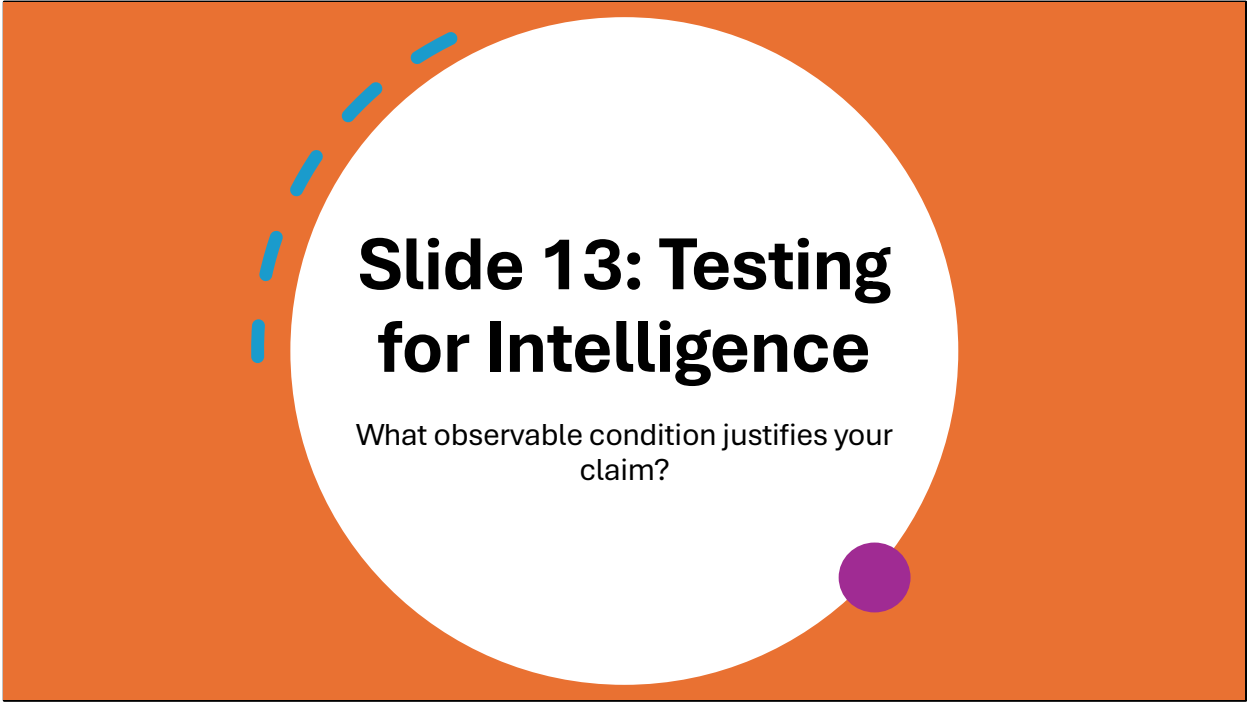


Oldowan artifacts are the oldest-known, simple stone tools, dating from 2.6 to 1.7 million years ago, representing the earliest, widely recognized lithic technology. Originating in East Africa, these tools (including flakes, cores, and choppers) were used by early hominins like Homo habilis to butcher meat, smash bones, and cut plants, marking a major, sustained, evolutionary shift.



Can non-humans be intelligent?

- **Are non-human primates intelligent?**
- Are birds intelligent?
- Are plants intelligent?
- Cells?

The slide features a solid orange background. A large white circle is centered on the slide. To the left of the circle, a dashed blue arc curves around its edge. A small solid purple circle is positioned on the right side of the white circle. The title "Slide 13: Testing for Intelligence" is written in bold black text inside the white circle. Below the title, the question "What observable condition justifies your claim?" is written in a smaller black font.

Slide 13: Testing for Intelligence

What observable condition justifies your
claim?

Crow solves puzzle



<https://www.youtube.com/shorts/dypodx8GU3c>

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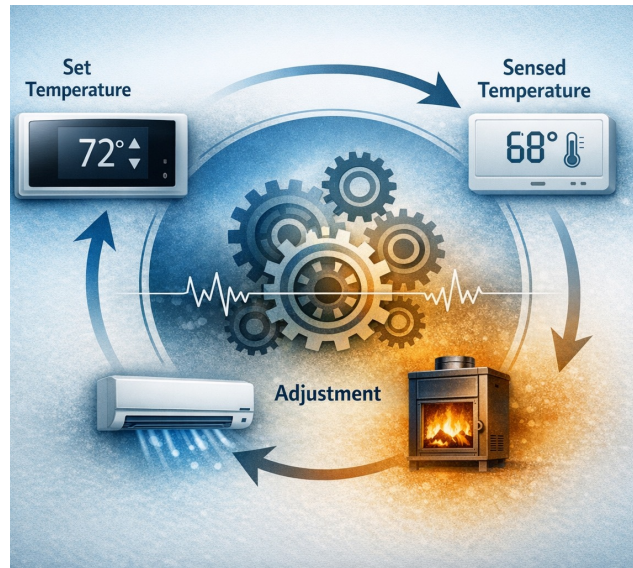
Murmuration

<https://www.youtube.com/shorts/FlbWQhI87cY>



<https://www.youtube.com/shorts/FlbWQhI87cY>

Is regulation Intelligent?



Moral Machine

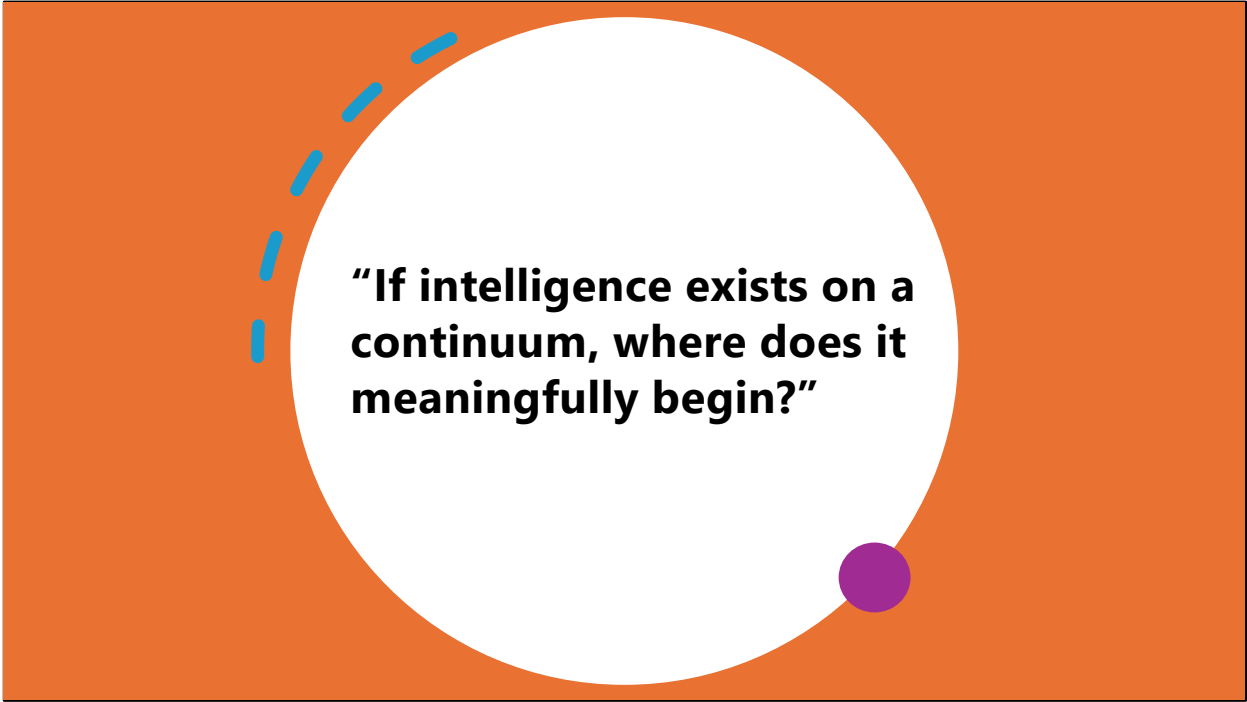
Where is the intelligence?
Where is the decision?
Where is the responsibility?

<https://www.moralmachine.net/>

<https://www.moralmachine.net/>



**If ethical behavior
requires intelligence,
what do we mean by
“intelligence”?**



"If intelligence exists on a continuum, where does it meaningfully begin?"

Origins of Theories of Intelligence

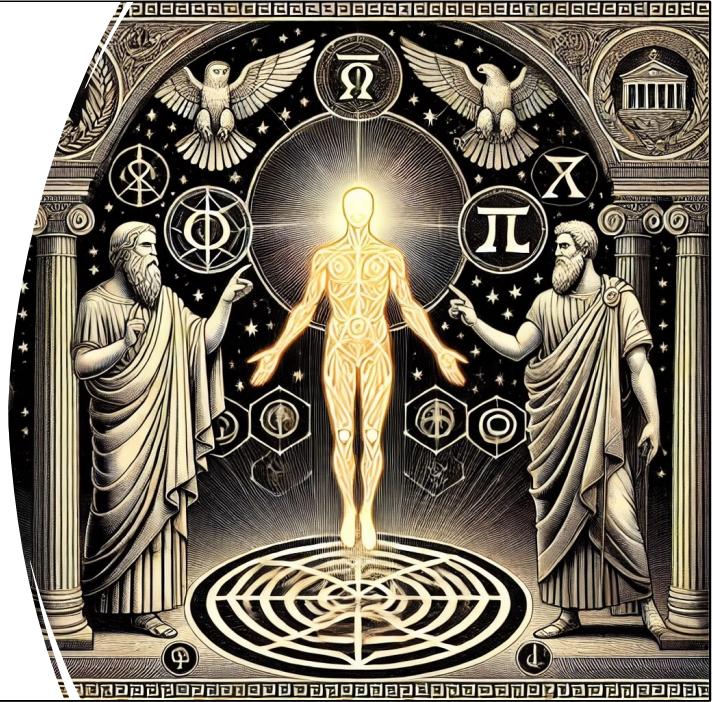
Intelligence as Divine Essence / Animating Force

- (2000 BCE)
- **Representative:** Plato (*Timaeus*); Hebrew Bible (*Genesis*)
- **Theory:** Intelligence is viewed as a "breath" or spirit provided by a cosmic or divine order. It is the animating force that distinguishes living beings from inert matter, making mind and spirit inseparable.



Intelligence as Rational Thought

- (500 BCE)
- **Representative:** Aristotle (*De Anima*, *Nicomachean Ethics*)
- **Theory:** Intelligence is the capacity for *logos*—the ability to use structured thought and logic to deliberate between alternatives and arrive at a rational choice.



Intelligence as Social Harmony

- (~500 BCE)
Representative: Confucius – *Analects*
- Intelligence as cultivated judgment (*zhi*)
- Intelligence as appropriate action within social roles
- Intelligence as ethical discernment sustaining social harmony



Intelligence as Insight and Alignment with Natural Order

- (~400 BCE)

Representative: Laozi – *Tao Te Ching*; Buddhist traditions – *Abhidharma* texts

- Intelligence as insight or awakened awareness
- Intelligence as perception of underlying patterns of reality •
- Intelligence as alignment with natural order rather than control



Intelligence as Ordered Moral Knowledge

- (400 CE)
- **Representative:** Augustine (*Confessions*); Aquinas (*Summa Theologiae*)
- **Theory:** Intelligence is the ability to recognize and align with a universal moral and theological order. "Knowing" is the wisdom to perceive "The Good" as the ultimate truth.



Intelligence as Mechanical Function

- (17th–18th Century)
- **Representatives:** René Descartes (*Treatise of Man*); Thomas Hobbes (*Leviathan*); early mechanical philosophers
- Mind compared to clockwork or mechanism
- Intelligence as structured process governed by rules
- Emergence of formal models of system behavior



Theory:

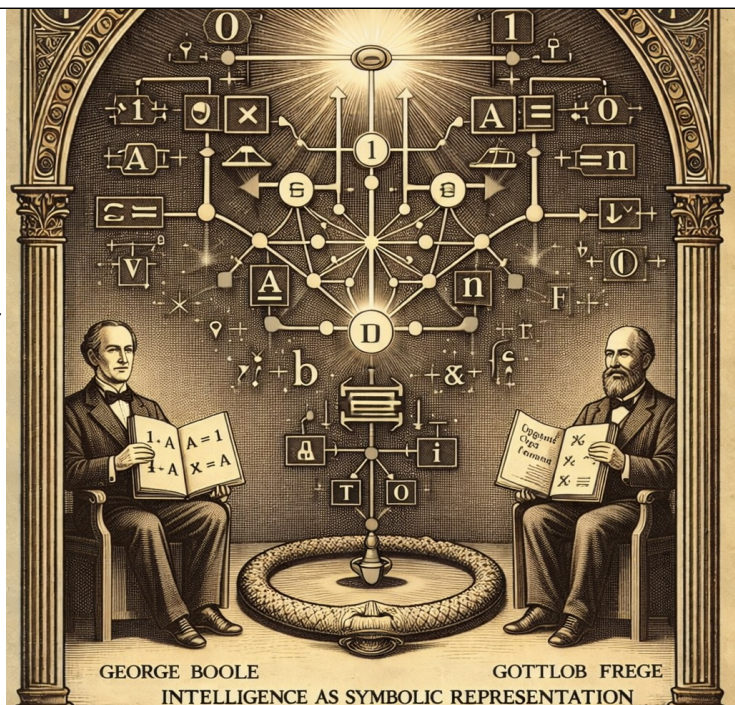
Intelligence is understood as the **operation of a physical mechanism** governed by deterministic laws. Mental activity is reduced to **motion, cause-and-effect relations, and rule-governed processes**, analogous to clocks, mills, or automata.

Thinking does not require insight or awareness; it is the **correct execution of internal mechanisms**. Intelligence is therefore measurable by **functional performance**, not by inner experience or moral understanding.

This conception marks a decisive shift: intelligence becomes **separable from life, meaning, and consciousness**, opening the conceptual path toward artificial and machine intelligence.

Intelligence As Symbolic Representation

- (19th–Early 20th Century)
- **Representative:** George Boole – *The Laws of Thought*
Frege – *Begriffsschrift*
- Intelligence as manipulation of symbolic representations
- Reasoning formalized through logic and abstraction •
- Foundations of computational and formal models of thought



Speaker Notes – Intelligence as Symbolic Representation

Intelligence reconceived as **manipulation of symbols**, not physical motion

Thought treated as **formal representation and rule-based transformation**

Intelligence located in:

- symbols
- syntax
- logical structure

Meaning treated as **secondary or external**

Reasoning becomes:

- abstract
- explicit
- formalizable

Thinking can be **written down, encoded, and executed**

Separation deepens:

- symbols vs world
- syntax vs semantics
- process vs understanding

Enables intelligence to be:

- standardized

evaluated independently of embodiment
implemented computationally

Establishes foundations for:

logic
computation
classical AI

Key tension to surface:

If symbols are manipulated correctly,
where does meaning come from?

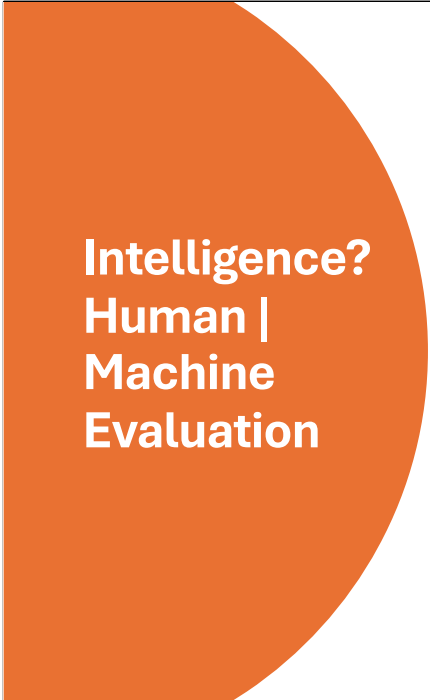
Roots of the Term

The Latin verb ***intelligere***:

•***inter*** — "between," "among"

•***legere*** — "to gather," "to select,"
"to choose," "to read"





Intelligence? Human | Machine Evaluation

Using the criteria generated earlier (Slide How do We recognize Intelligence), determine whether each condition applies to:

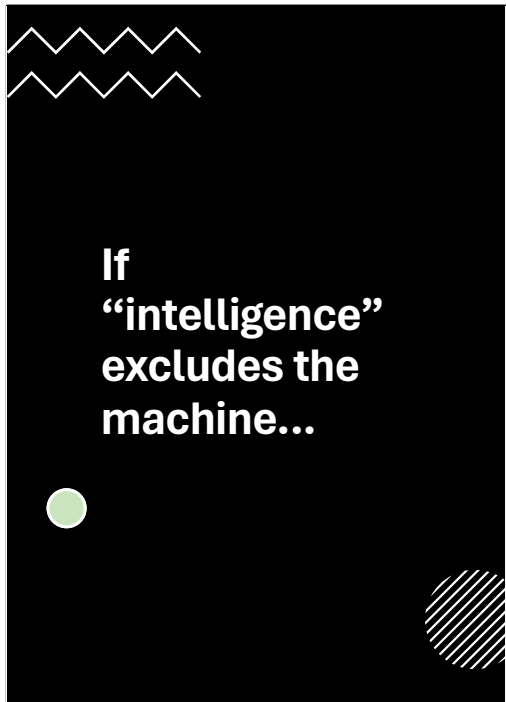
- Human
- Machine
- Both





If internal states are not **observable**, should intelligence be defined behaviorally?





**If
“intelligence”
excludes the
machine...**

Premise. If machines are not intelligent, then the qualities humans and machines share need a different name.

Task. Name the shared qualities (start from the “Both” column on slide 38). Then propose a term that fits them — and only them.

Test. Does your term avoid “intelligence” and its synonyms? Does it still apply to the crow and the murmuration?

Proposed term: _____

If no better term survives, the term we need is the one we removed.

Purpose of this slide. This is the pivot slide of the session — and it is deliberately a rhetorical trap. Up to this point the class has (a) built a working test framework for intelligence, (b) stress-tested it against crows, murmurations, plants, cells, regulation, and the Moral Machine, (c) walked through eight historical theories, and (d) landed on the Latin root *intelligere* = *inter* + *legere* (“to choose between, to gather, to select”). Slide 38 just forced a Human / Machine / Both judgment, and slide 39 asked whether intelligence should be defined behaviorally. This slide cashes that in. The premise is binary: if machines are not intelligent, then the qualities humans and machines genuinely share need a different name. So — propose one.

The reveal (do not say this until step 5). Every candidate term the class proposes will collapse under examination. Synonyms (cognition, mind, reasoning, thinking) smuggle “intelligence” back in. Narrower terms (computation, processing) feel evacuated of what we actually want to name. Broader terms (adaptation, behavior) lose their grip on what’s distinctive. The class will find that no other term survives both tests cleanly. That leaves two and only two honest conclusions: (1) the premise was wrong — machines ARE intelligent, and our working definition needs to expand; or (2) “intelligence” is

itself the right term, and it applies to humans and machines both. Either way, the move the class resisted on slide 38 is now forced. This is the point of the exercise.

How to run it (about 8–10 minutes).

1. Read the premise aloud. Pause. Ask: “Do you accept this premise, even hypothetically?” Take a quick show of hands — this matters, because students who reject the premise will resist the task.
2. Surface the observation. Point back to the “Both” column from the Human / Machine evaluation on slide 38. Ask the class to name two or three concrete capabilities that landed there (e.g., pattern recognition, goal-directed selection, error correction, adaptation to feedback).
3. Pose the naming task. Give students 2–3 minutes in pairs to propose a single term for the shared capability set. Require a one-sentence definition with the term.
4. Collect 4–6 proposed terms on the board. Do not evaluate them yet — just collect.
5. Apply two tests to each candidate: (i) Does it avoid “intelligence” and its synonyms? (ii) Does it still apply to the crow and the murmuration? Cross out terms that fail either test. When the board is empty — or nearly empty — surface the reveal: if no better term survives, the term we need is the one we removed, and the premise that machines are not intelligent has to be revisited.

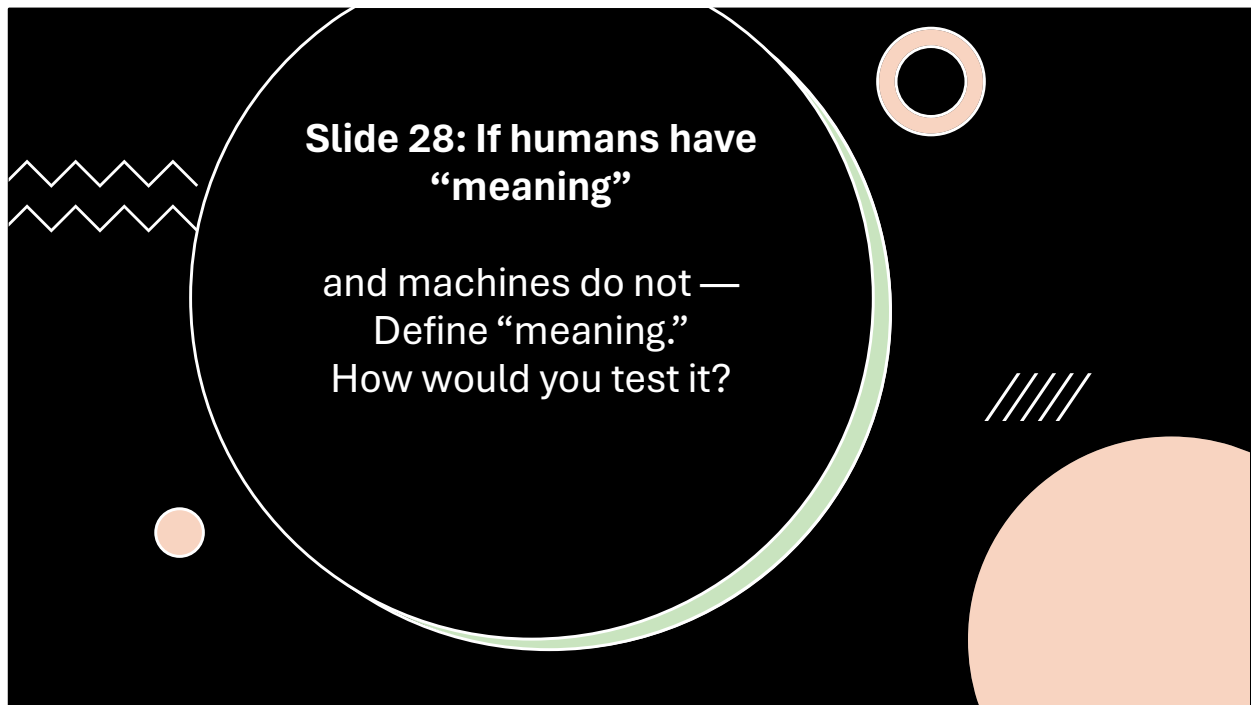
Candidate terms to expect (and what they reveal).

- Processing / computation — honest but evacuates normativity; sets up slide 42’s jump to agency and responsibility.
- Selection / choosing-between — cashes in the inter-legere etymology from slide 37; the deck’s implicit answer.
- Adaptation / learning — broad enough to include the murmuration and the crow; tests whether students will accept distributed cases.
- Functional cognition — academic, hedged; useful to flag as a placeholder that defers the question.
- Agency — students will reach for this. Resist letting them collapse it here; agency is slide 42’s territory and depends on what we name today.

The move you are setting up. Whatever term the class lands on becomes the seed noun for the “provisional functional definition” due as homework on slide 43. Slide 41 then asks the symmetric question about meaning; slide 42 ladders from this term to agency, responsibility, and accountability. If the class lands on “selection” or a near-synonym, you have a clean bridge to inter-legere; if they land on “computation,” you have a sharper conflict to mine on slide 42.

Common student moves to watch for.

- Smuggling “intelligence” back in via synonyms (cognition, mind, thinking). Gently rule these out — the point is to find a term that does not presuppose the one we removed.
- Rejecting the premise (“but machines aren’t excluded”). Acknowledge it and ask them to play along hypothetically; the exercise still works.
- Jumping straight to ethics (“responsibility”). Park it — slide 42 is built for this.



Purpose of this slide. This is the symmetric move to slide 40. There the class confronted whether the qualities humans and machines share need a name other than “intelligence.” Here the move flips: the class is invited to defend the asymmetry — humans have “meaning,” machines do not — and is then asked to operationalize the very thing they used to draw the line. This slide is also a deliberate trap. Just as no honest synonym for intelligence survived slide 40, no honest test for “meaning” will survive this one without either (a) collapsing into behavior (which machines can also produce), or (b) appealing to inner states we cannot observe (which slide 39 already ruled out as a basis for definition).

The reveal (do not say this until step 5). Every candidate definition of “meaning” will land in one of three places: behavioral (response to context, appropriate action, sensitivity to consequence — all of which a machine can be shown to do); phenomenal (subjective experience, what-it-is-like, qualia — none of which can be tested from the outside); or normative (mattering, caring, valuing — which only pushes the question one word over). The class is forced to choose: either “meaning” has a behavioral test, in which case machines can pass it; or it has no test at all, in which case it cannot do the work of excluding machines. Either way, the asymmetry the slide started with cannot survive its own evaluation criterion.

This is exactly the move slide 40 made for intelligence, repeated for meaning.

How to run it (about 8–10 minutes).

1. Read the conditional aloud and pause. Ask: “Do you accept the premise — that humans have meaning and machines do not?” Take a quick show of hands. Mark the room: confident yes, confident no, unsure.
2. Pose the definitional task. Give students 2–3 minutes in pairs to propose a one-sentence definition of “meaning” that the class can use today. Require the definition to be specific enough that someone else could apply it.
3. Pose the test task. Same pairs, 2 more minutes: how would you test for the presence of “meaning” in a given system? Require an observable procedure, not a feeling.
4. Collect 4–6 definition / test pairs on the board. Do not evaluate them yet.
5. Apply three pressures to each pair: (i) Could a large language model pass this test? (ii) Could a non-human animal pass it? (iii) Could a comatose human pass it? Watch which definitions survive all three and which split apart. Surface the reveal when the board collapses.

Candidate definitions to expect (and what they reveal).

- Reference / aboutness — “a state that is about something else.” Behaviorally testable; machines pass trivially. Fails to exclude.
- Understanding — students will reach for Searle’s Chinese Room intuition. Useful, but it relocates the question to “understanding” without defining it.
- Caring / mattering — relocates again, this time to value. Sets up slide 42’s ladder cleanly.
- Subjective experience / qualia — the honest phenomenal answer; admits no external test. Worth surfacing precisely because it shows where the asymmetry actually lives — and why it cannot do public work.
- Use in a context / communication — Wittgensteinian. Machines pass. Useful pivot if you want to push toward language and convention.

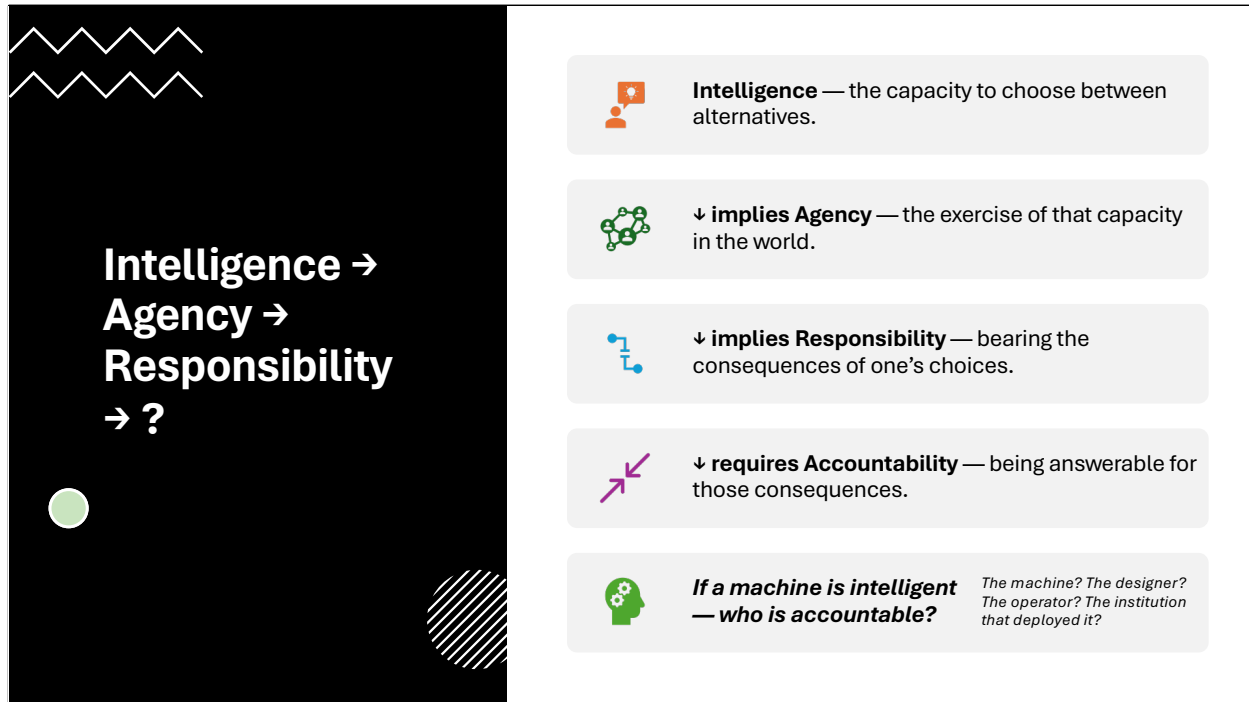
The move you are setting up. Slide 42 is the ladder: Intelligence → Agency → Responsibility → Accountability. Slide 40 destabilized the first rung; this slide destabilizes the cluster of properties (meaning, understanding, caring) we usually slip under the second. Whichever way the class lands — behavioral and inclusive, or phenomenal and untestable — slide 42 inherits the consequences. The homework on slide 43 then asks students to write a provisional functional definition; the two failed asymmetries (intelligence on slide 40, meaning here) tell them why a functional definition is necessary in the first place.

Common student moves to watch for.

- Defining meaning with another undefined term (purpose, soul, understanding,

consciousness). Push back gently — require an observable handle.

- Appealing to private experience to settle a public test. Acknowledge the honesty of the position, then note what it costs: it cannot do the work of excluding anything.
- Jumping to ethics (“meaning means moral standing”). Park it for slide 42 — but flag that the jump is being made.



Slide 30: Homework

- Research two definitions of intelligence.
- Do they require consciousness?
- Implications for agency and responsibility.
- Propose a provisional functional definition.